

Navigating the Privacy-Utility-Communication Trilemma: A Systematic Review of Adaptive and Hybrid Mechanisms in Differentially Private Federated Learning

Federated Learning (FL) mitigates data silo challenges but remains vulnerable to gradient leakage and membership inference attacks, necessitating the integration of Differential Privacy (DP) for rigorous guarantees [143]. However, this integration creates a complex "Privacy-Utility-Communication" trilemma, where static noise injection often degrades model convergence, particularly in non-IID settings [221]. This review synthesizes recent advancements transitioning from uniform perturbation to adaptive, context-aware mechanisms. We analyze layer-wise and time-adaptive strategies that dynamically allocate privacy budgets to optimize the trade-off between security and accuracy [22] [174]. Furthermore, we examine hybrid architectures combining DP with cryptographic primitives like Secure Aggregation and the Shuffle Model to achieve central-model utility without trusted curators [1] [30]. The review also addresses emerging frontiers, including the tension between privacy and fairness, heterogeneous client requirements, and efficient unlearning protocols [11] [45]. Finally, we highlight critical gaps in standardized benchmarking, noting that naive hybrid combinations can cause severe convergence failures [PrivacyBench: Privacy Isn't Free in Hybrid Privacy-Preserving Vision Systems]. We conclude that future research must prioritize holistic, adaptive frameworks that reconcile mathematical privacy guarantees with the practical constraints of heterogeneous, resource-limited distributed systems.

Introduction to Privacy-Preserving Federated Learning Paradigms

The evolution of machine learning architectures has shifted decisively from centralized data repositories to decentralized paradigms, driven by the ubiquity of mobile devices, stringent regulatory frameworks, and growing public concern over data sovereignty. Federated Learning (FL) emerged as a transformative solution to these "data silo" challenges, enabling collaborative model training while keeping raw data localized on client devices [36]. By exchanging only model parameters or gradients rather than sensitive datasets, FL ostensibly mitigates the risks associated with central data collection [282]. However, the assumption that local data retention equates to absolute privacy has been rigorously challenged. Recent studies demonstrate that FL systems remain vulnerable to sophisticated inference attacks, where adversaries can reconstruct private training data or infer membership information by analyzing shared gradient updates [143]. Consequently, FL alone is insufficient for robust privacy protection, necessitating the integration of formal privacy guarantees such as Differential Privacy (DP) [12].

The integration of DP into FL introduces a complex "Privacy-Utility-Communication" trilemma. DP provides a mathematically rigorous definition of privacy that limits information leakage about any single individual, regardless of the adversary's computational power [12]. When applied to FL, typically through mechanisms like DP-SGD, noise is injected into gradients or model updates to obscure individual contributions [214]. While this effectively counters gradient leakage and membership inference attacks, it invariably degrades model utility and can impede convergence, particularly in non-IID data settings common in real-world deployments [221]. Furthermore, the addition of noise increases the variance of updates, which may require more communication rounds to achieve target accuracy, thereby exacerbating bandwidth constraints on edge devices [242]. This trade-off is central to current research, with numerous works aiming to optimize the balance between privacy budgets, model performance, and communication efficiency [1]. The interplay between communication reduction techniques and DP mechanisms presents additional challenges, as compression can interact unpredictably with privacy noise [149].

A critical divergence in the literature concerns the architectural approach to privacy: centralized versus decentralized aggregation. Traditional FL relies on a central server to aggregate updates, often employing Central DP (CDP) where noise is added by a trusted curator or via secure aggregation protocols. However, the trust assumption regarding the central server is a significant vulnerability. In response, Local DP (LDP) approaches have gained traction, where clients perturb their own updates before transmission, eliminating the need to trust the server [25]. Yet, LDP typically incurs a higher utility cost due to the lack of noise cancellation inherent in CDP [Local and Central Differential Privacy for Robustness and Privacy in Federated Learning]. Interestingly, recent theoretical and empirical evidence suggests that Decentralized Federated Learning (DFL), which utilizes peer-to-peer communication, may offer superior privacy properties compared to centralized models. Contrary to findings suggesting no additional benefit, some studies argue that DFL theoretically upper-bounds privacy leakage lower than centralized FL because adversaries observe only differences of local gradients rather than the gradients themselves [273]. This has spurred the development of hybrid frameworks that combine the statistical noise of DP with cryptographic protocols like Secure Aggregation (SA) and Multi-Party Computation (MPC) to achieve the utility benefits of CDP with the trust assumptions of LDP [4].

The landscape of DP-FL is further complicated by the heterogeneity of client requirements and system constraints. Uniform privacy budgets often fail to account for varying data sensitivities across clients, leading to suboptimal utility for those with less sensitive data. To address this, researchers have proposed Heterogeneous DP (HDP) and adaptive mechanisms that dynamically adjust noise levels based on gradient sensitivity, training rounds, or client contribution [35]. For instance, adaptive noise injection and layer-wise perturbation strategies have shown promise in reducing the overall noise burden while maintaining robust defense against reconstruction attacks [22]. Additionally, the tension between robustness against poisoning attacks and privacy preservation remains unresolved, as defenses optimized for one objective may inadvertently weaken the other [144]. Addressing these contradictions and synthesizing a coherent framework for hybrid privacy-preserving FL is essential for the next generation of trustworthy distributed intelligence, especially given that existing surveys often lack a unified taxonomy addressing the interplay between different privacy models and neighborhood levels [2]. Furthermore, the practical deployment of these systems is hindered by the difficulty of verifying server-side DP guarantees and coordinating training across highly heterogeneous devices [247].

The Evolution from Centralized to Decentralized Architectures

The paradigm shift from centralized deep learning to decentralized architectures represents a fundamental reimagining of machine learning, driven primarily by the imperative to resolve the "data island" problem while adhering to stringent privacy regulations. Traditional centralized approaches necessitate the aggregation of raw data into a single repository, creating significant vulnerabilities regarding data sovereignty and exposure. In contrast, Federated Learning (FL) emerges as a distributed alternative where raw data remains localized on client devices, and only model parameters or gradient updates are shared with a central aggregator [282]. This architectural evolution theoretically mitigates the risk of direct data leakage; however, it introduces new attack surfaces where sensitive information can be inferred from the shared updates themselves. Consequently, the mere locality of data is insufficient for robust privacy, necessitating the integration of formal privacy definitions such as Differential Privacy (DP) to protect against sophisticated inference attacks [218].

While FL prevents the direct transfer of raw datasets, research has demonstrated that adversaries can still reconstruct training data or perform membership inference attacks by analyzing the shared model updates [51]. This vulnerability underscores the limitation of relying solely on decentralized data storage as a privacy mechanism. To address this, the integration of DP into FL has become a standard approach, providing rigorous, quantifiable guarantees against privacy breaches regardless of the adversary's computational power [12]. However, the application of DP in decentralized settings introduces a complex trade-off between privacy preservation and model utility. The injection of noise required to satisfy DP constraints often degrades model accuracy, particularly in deep neural networks with high-dimensional parameter spaces [159]. Furthermore, empirical studies suggest that theoretical privacy bounds can be overly conservative, with actual leakage often being lower than predicted, yet still significant enough to warrant careful calibration [86].

To mitigate the utility loss associated with uniform noise injection, recent literature proposes adaptive and hierarchical mechanisms. For instance, layer-wise adaptive noise injection strategies leverage the insight that different neural network layers contribute unevenly to model utility and privacy leakage, allowing for optimized noise distribution that preserves accuracy while maintaining strong privacy guarantees [22]. Similarly, approaches utilizing sparsified model perturbation reduce the dimensionality of the perturbation by sparsifying local models before noise addition, effectively improving the privacy-utility trade-off [281]. Additionally, the concept of heterogeneous differential privacy acknowledges that clients may have varying privacy requirements, proposing algorithms that personalize privacy budgets to enhance overall system performance without compromising individual constraints [122]. Other innovations include time-adaptive privacy spending, where clients save privacy budgets in early training rounds to expend more on fine-grained features in later rounds, further optimizing the utility-privacy balance [174].

The evolution of these architectures also extends to the aggregation mechanism itself, challenging the "honest-but-curious" server assumption. Early FL models relied on a trusted central server, but findings indicate that servers can potentially isolate and analyze individual updates even within secure aggregation frameworks, especially through Sybil attacks or disaggregation techniques [Reconstructing Individual Data Points in Federated Learning Hardened with Differential Privacy and Secure Aggregation]. This has spurred the development of decentralized trust models and advanced cryptographic techniques. Frameworks introducing decentralized consensus and threshold functional encryption prevent disaggregation attacks by malicious or curious aggregators, thereby distributing trust across the network [241] [296]. Moreover, fully decentralized learning paradigms eliminate the central server entirely by utilizing virtual nodes and correlated noise to obscure the origin of model updates, offering enhanced resistance to privacy attacks in peer-to-peer settings [60] [95].

Despite these advancements, contradictions and challenges remain regarding the practical deployment of these systems. Some studies suggest that decentralized optimization inherently provides better privacy bounds than centralized FL due to the obfuscation of gradient differences over successive iterations [273]. Conversely, other research indicates that without careful calibration, the cumulative privacy loss over multiple communication rounds can render theoretical guarantees vacuous in practice. The integration of secure aggregation with DP further complicates the landscape; while secure aggregation prevents the server from viewing individual updates, it does not inherently provide formal DP guarantees unless specific conditions regarding the randomness of the aggregate are met, which are rarely satisfied in practice without additional noise [79]. Consequently, hybrid approaches combining homomorphic encryption, secure

multi-party computation, and DP are increasingly viewed as necessary to address the full spectrum of threats, from external eavesdroppers to colluding data holders [91]. The ongoing evolution from centralized to decentralized architectures thus reflects a continuous negotiation between computational efficiency, communication overhead, and the rigorous mathematical guarantees required to ensure true privacy in collaborative learning environments.

Fundamental Concepts and Threat Models

Differential Privacy (DP) serves as the rigorous mathematical foundation for privacy-preserving Federated Learning (FL), offering quantifiable guarantees against adversaries with arbitrary background knowledge by ensuring that the inclusion or exclusion of a single record does not significantly alter the output distribution [12]. In the federated context, this definition bifurcates into distinct models based on trust assumptions regarding the aggregator and the locus of noise injection. Local Differential Privacy (LDP) operates under the strictest threat model, requiring clients to perturb their data or gradients locally before transmission, thereby ensuring privacy even against a malicious server [25]. Conversely, Central Differential Privacy (CDP) assumes a trusted curator who aggregates raw updates and adds noise only to the final result, offering superior utility but necessitating significant trust in the central entity [28]. Research indicates that while both LDP and CDP can defend against backdoor and membership inference attacks, neither fully mitigates property inference, highlighting specific gaps in current threat models [Local and Central Differential Privacy for Robustness and Privacy in Federated Learning].

The tension between these extremes has catalyzed the emergence of the Shuffle Model, which introduces an intermediate shuffler to anonymize perturbed messages before they reach the aggregator. This architecture leverages privacy amplification by shuffling, allowing systems to achieve privacy guarantees approaching those of the central model while maintaining the distrust-free nature of the local model [30]. Recent advancements demonstrate that the shuffle model can significantly tighten privacy bounds through mechanisms like subsampling and gradient sparsification, effectively bridging the utility gap between LDP and CDP [123]. However, the efficacy of these models relies heavily on the specific threat model adopted. Standard analyses often assume an "honest-but-curious" server that follows the protocol but attempts to infer private information from observed updates [43]. Yet, empirical studies suggest that theoretical LDP bounds can be overly conservative, and instantiated adversaries in FL may not always reach the worst-case theoretical limits, prompting discussions on relaxing privacy parameters for practical utility [229].

Beyond the honest-but-curious assumption, the landscape of threats extends to malicious Byzantine agents capable of deviating arbitrarily from the protocol to poison models or disrupt convergence. Addressing these dual threats of privacy leakage and Byzantine robustness presents a complex challenge, as the noise added for DP can sometimes interfere with robust aggregation rules designed to filter out malicious updates [245]. Innovative frameworks have been developed to decouple these concerns, utilizing techniques such as client momentum to average updates over time, thereby exposing malicious perturbations while maintaining differential privacy through secure aggregation [67]. Furthermore, the integrity of the shuffling process itself is critical; if the shuffler is compromised, the privacy amplification collapses. To mitigate this, recent protocols incorporate cryptographic primitives like secret-shared shuffles and lightweight integrity checks to ensure security against malicious adversaries without sacrificing communication efficiency [123]. Similarly, frameworks like RAIN introduce robust aggregation in noise to detect selective dropping and tampering, ensuring verifiability alongside privacy [52].

The distinction between adding noise at the user level versus the aggregator level fundamentally alters the system's resilience to gradient leakage attacks. While CDP offers better utility, it remains vulnerable if the server is compromised, whereas LDP provides end-to-end protection at the cost of higher noise variance [Local and Central Differential Privacy for Robustness and Privacy in Federated Learning]. The dimensionality of model updates exacerbates the noise required in LDP, leading to frameworks that employ dimension selection or correlated noise to maintain utility without compromising privacy guarantees [297]. The integration of secure aggregation (SecAgg) further complicates the threat model; while SecAgg prevents the server from viewing individual updates, it does not inherently provide DP guarantees without additional noise, as the aggregate itself may leak information about individual datasets over multiple rounds [218]. Theoretical analysis confirms that conditions under which SecAgg alone provides DP are practically unlikely, necessitating additional noise injection [79]. Consequently, modern approaches often hybridize these techniques, utilizing distributed DP combined with secure aggregation to ensure that no single compromised server or channel can reveal individual contributions [4].

Ultimately, the selection of a DP model and the corresponding threat assessment must account for the heterogeneity of data and the specific capabilities of potential adversaries. Algorithms like DP-SCAFFOLD address data heterogeneity by controlling user-drift while maintaining formal privacy guarantees against honest-but-curious servers [43]. Meanwhile, advanced accounting methods, such as those based on Rényi DP or ϵ -DP, allow for tighter tracking of

privacy loss across iterations, enabling more efficient use of the privacy budget in complex, heterogeneous environments [6]. As the field evolves, the focus shifts towards adaptive mechanisms that dynamically adjust noise levels based on gradient importance or training stages, aiming to optimize the delicate balance between robustness, privacy, and model utility [84].

The Privacy-Utility-Communication Trilemma

The integration of Differential Privacy (DP) into Federated Learning (FL) establishes a fundamental tension often described as a trilemma involving privacy guarantees, model utility, and communication efficiency. While FL inherently protects data by keeping it local, the exchange of model parameters or gradients remains vulnerable to inference attacks, necessitating the addition of noise to satisfy rigorous DP standards [12]. However, this noise injection directly degrades model accuracy and convergence speed, creating an immediate conflict between privacy and utility [214]. This challenge is exacerbated when considering communication constraints; reducing bandwidth through compression or quantization can inadvertently alter privacy guarantees or introduce additional variance that hampers convergence [149]. Consequently, achieving a viable system requires navigating these competing objectives, where enhancing one dimension often necessitates a sacrifice in another.

A prominent strategy to address this trilemma is the "belt and braces" approach, which combines FL with DP to offer comprehensive protection against both honest-but-curious servers and external adversaries [1]. This dual-layered defense, however, introduces significant optimization challenges. Traditional methods that rely solely on adding local noise often result in unacceptable drops in training accuracy, while combining Secure Multi-Party Computation (SMPC) with DP improves accuracy but incurs prohibitive communication and computation overheads [8]. To mitigate these issues, recent research has explored adaptive mechanisms that dynamically adjust noise levels based on training progress or feature importance. For instance, adaptive noise injection schemes that decrease perturbation as the model converges, or those that prioritize noise addition to less critical model parameters, have shown promise in preserving utility without compromising privacy budgets [293] [227]. Furthermore, time-adaptive privacy spending allows clients to conserve privacy budgets in early training rounds for coarse-grained features, reserving higher precision for later stages where fine-grained learning is critical [174].

The communication aspect of the trilemma presents unique opportunities and contradictions, particularly in wireless environments. In over-the-air FL (AirFL), the inherent channel noise of wireless transmissions acts as a "frenemy"; while it degrades signal quality, it also provides a natural source of randomness that can be harnessed for DP, potentially offering "free privacy" without injecting artificial noise [151] [130]. However, this benefit is not universal. In Multiple-Input Multiple-Output (MIMO) systems, the use of multiple antennas at the base station can amplify privacy leakage if separate receive combining is employed, rendering reliance on channel noise insufficient and necessitating device-side privacy mechanisms [161]. This contradiction highlights that the efficacy of leveraging communication noise is highly dependent on the specific network architecture and transceiver design.

To further reconcile the trade-offs, researchers have investigated the synergy between quantization and privacy. Quantization, primarily used to reduce communication overhead by compressing model updates, inherently introduces randomness that can contribute to DP guarantees. Studies indicate that lower quantization bit levels can actually enhance privacy protection, suggesting a potential alignment between communication efficiency and privacy rather than a strict trade-off [The Effect of Quantization in Federated Learning: A Renyi Differential Privacy Perspective] [270]. Novel frameworks like DPSFL utilize differentially private sketches to compress gradients while adding noise to the sketches themselves, thereby addressing both communication inefficiency and privacy leakage simultaneously [183]. Similarly, adaptive quantization strategies that adjust bit-lengths based on client data entropy have demonstrated significant reductions in communicated data while maintaining competitive accuracy under DP constraints [172].

Despite these advancements, the trilemma remains a persistent hurdle, particularly in heterogeneous settings. The presence of non-IID data distributions complicates the convergence of DP-FL, often requiring more communication rounds to achieve target accuracy, which in turn accumulates privacy loss [43]. Approaches such as DP-SCAFFOLD attempt to correct user drift in heterogeneous environments to improve the privacy-utility trade-off, yet the fundamental tension persists. Moreover, the interplay between fairness, privacy, and utility adds another layer of complexity; ensuring fair performance across diverse clients while adhering to strict DP budgets often requires sacrificing overall model accuracy or increasing communication overhead to accommodate personalized noise levels [93] [11]. Ultimately, while the "belt and braces" approach provides robust security, its practical deployment demands sophisticated optimization of noise variance, communication protocols, and aggregation strategies to prevent the system from becoming impractical due to slow convergence or excessive bandwidth consumption.

Scope and Motivation for Systematic Review

The rapid proliferation of Federated Learning (FL) has established it as a cornerstone paradigm for privacy-preserving machine learning, enabling collaborative model training across distributed clients without the exchange of raw data [40]. Despite its theoretical promise of data localization, the FL framework is not inherently immune to privacy breaches, as shared model updates remain vulnerable to sophisticated inference attacks that can reconstruct sensitive user information [218; 51]. Consequently, the integration of Differential Privacy (DP) has become the de facto standard for providing rigorous, mathematical guarantees against such adversaries [1]. However, the literature reveals a significant fragmentation regarding the taxonomy of adaptive mechanisms and their application across diverse domains, creating a critical need for a holistic review that synthesizes recent advancements in resolving the persistent privacy-utility deadlock.

A primary source of fragmentation lies in the heterogeneous approaches to managing the trade-off between privacy guarantees and model utility. Traditional implementations often rely on fixed privacy budgets and static clipping norms, which frequently result in suboptimal convergence or excessive noise injection that degrades model performance [88]. To address this, a growing body of work has shifted toward adaptive mechanisms, yet these solutions are often siloed within specific contexts. For instance, researchers have proposed dynamic clipping strategies that adjust norms based on multi-objective optimization frameworks, effectively balancing privacy loss with accuracy requirements [88]. Similarly, time-adaptive privacy spending strategies have been introduced to allocate privacy budgets non-uniformly across training rounds, recognizing that early rounds require less signal-to-noise ratio for coarse feature learning compared to later rounds [174]. Furthermore, adaptive methods based on cosine similarity and scoring functions have been developed to adjust privacy budgets dynamically according to accuracy, loss, and dataset characteristics, thereby reducing unnecessary privacy expenditure while maintaining security [35]. Despite these individual advancements, there is a lack of unified understanding regarding how these adaptive strategies interact with different optimization schemes and data distributions, leading to inconsistent deployment guidelines.

The complexity is further compounded by the emergence of conflicting objectives beyond the binary privacy-utility trade-off, specifically involving fairness, personalization, and efficiency. Recent studies indicate an inherent tension between strict privacy protection and the ability to ensure fairness across demographic groups, where stricter privacy constraints can fundamentally limit the detection and correction of biases [277]. This tension is particularly acute in Personalized Federated Learning (PFL), where the interplay between privacy noise, model convergence, and performance distribution fairness remains underexplored, with few works addressing the joint optimization of these three factors [93]. Additionally, the rise of Heterogeneous Differential Privacy (HDP), where clients select individual privacy budgets, introduces new vulnerabilities; specifically, ϵ -aware aggregation can inadvertently expose structural patterns in gradients, enabling inference attacks even when local DP is applied [65]. This highlights a gap in current literature where privacy mechanisms designed for homogeneous settings fail to account for the nuanced risks introduced by privacy heterogeneity and client-specific constraints [120].

Moreover, the application domains of privacy-preserving FL exhibit distinct challenges that are often treated in isolation, preventing the development of generalizable solutions. In wireless environments, foundational questions regarding the precise characterization of privacy loss and convergence under non-convex objectives remain open, with existing analyses often relying on restrictive assumptions [71]. In the realm of Federated Unlearning, the necessity to erase specific client data introduces unique privacy risks related to stored historical updates, necessitating novel budget allocation strategies that balance unlearning efficiency with privacy preservation [45]. Similarly, in resource-constrained IoT and edge computing scenarios, the intersection of compression, communication efficiency, and privacy creates a multi-dimensional optimization problem that standard DP mechanisms often fail to address holistically [42; 289]. The literature also points to contradictions regarding the efficacy of Secure Aggregation (SA) when combined with DP; while some theoretical works suggest SA can provide DP without added noise under specific Gaussian conditions, practical evaluations indicate that additional noise is almost invariably required to achieve meaningful worst-case guarantees [79].

The current landscape is further fragmented by the lack of standardized evaluation metrics and the disparate treatment of system-level constraints. Benchmarks have revealed that hybrid configurations, such as combining FL with DP, can lead to severe convergence failures and drastic accuracy drops in certain vision systems, whereas other combinations

like FL with Secure Multi-Party Computation maintain performance, suggesting that privacy techniques cannot be composed arbitrarily without systematic guidance [PrivacyBench: Privacy Isn't Free in Hybrid Privacy-Preserving Vision Systems]. Furthermore, the limits of DP in continuous data collection and its misuse in record-level data release raise fundamental questions about the applicability of standard DP definitions in long-term FL deployments [178]. Given these disjointed advancements, contradictory findings on the necessity of noise addition, and the emerging multi-objective challenges involving fairness, unlearning, and heterogeneity, there is a compelling motivation for a systematic review. Such a review is essential to synthesize the fragmented literature, reconcile theoretical bounds with empirical realities, and establish a coherent framework for designing adaptive, multi-objective optimization strategies that can effectively resolve the privacy-utility deadlock in next-generation federated learning systems.

Theoretical Frameworks and Privacy Models

The theoretical landscape of Differentially Private Federated Learning (DP-FL) has undergone a significant paradigm shift, evolving from a binary distinction based on server trustworthiness to a rigorous taxonomy grounded in mathematical definitions, neighborhood levels, and specific federated scenarios. Early frameworks primarily categorized approaches based on the presence of a central curator, distinguishing between Central Differential Privacy (CDP), which assumes a trusted aggregator, and Local Differential Privacy (LDP), which operates under the assumption of an untrusted server [2]. However, this dichotomy proves insufficient for modern composite architectures, such as cross-silo settings where clients may manage local servers, necessitating a more granular classification based on the specific guarantees of privacy models and the objects they protect [2]. Contemporary taxonomies now delineate privacy protections across sample-level and client-level neighborhood definitions, recognizing that the addition or removal of an entire client's dataset constitutes a distinct neighboring relation compared to single-record changes [2]. This distinction is critical for determining the sensitivity of aggregation functions and the magnitude of noise required to satisfy (ϵ, δ) -DP guarantees in diverse federated environments [11].

At the core of these frameworks lies the adaptation of differential privacy's mathematical formalism to address specific threat models in distributed learning. While LDP offers strong privacy notions by requiring data owners to perturb inputs before transmission, it often incurs significant utility costs due to the high volume of noise needed for high-dimensional deep learning models [25]. To bridge the gap between the high utility of CDP and the strong trust assumptions of LDP, the shuffle model has emerged as a pivotal theoretical advancement. By leveraging a trusted shuffler to anonymize user updates, the shuffle model achieves a privacy amplification effect that allows for curator-level accuracy without requiring a fully trusted analyzer [30]. Theoretical analyses demonstrate that protocols employing subsampling and gradient sparsification within the shuffle model can effectively handle large model dimensions while maintaining robust privacy guarantees [30]. Furthermore, recent developments in Federated ϵ -Differential Privacy provide a unified framework based on Gaussian differential privacy, offering tighter privacy accounting for record-level guarantees against adversaries in iterative settings [37].

The interplay between privacy definitions and optimization dynamics introduces complex challenges, particularly in heterogeneous data environments characterized by non-IID distributions. Traditional approaches like DP-SGD, which involve per-sample gradient clipping and noise addition, can severely degrade convergence under such conditions [43]. To mitigate client drift and improve utility, advanced frameworks have introduced mechanisms such as control variates, as seen in DP-SCAFFOLD, which outperforms standard DP-FedAvg by correcting user drift while maintaining formal privacy bounds [43]. Additionally, the integration of fairness constraints necessitates a re-evaluation of privacy accounting, as fairness-aware gradient clipping and loss uploading consume additional privacy budgets, requiring adaptive strategies to balance fairness, privacy, and utility simultaneously [11]. In scenarios with heterogeneous privacy requirements, where clients select individual privacy budgets, ϵ -aware aggregation can inadvertently expose structural patterns in gradients; thus, new middleware defenses like IntraShuffler propose parameter-level shuffling within privacy-compatible buckets to disrupt persistent gradient structures while maintaining utility [65].

Recent theoretical advancements also challenge the necessity of explicit noise addition under specific conditions, particularly regarding Secure Aggregation (SA). While SA prevents the server from viewing individual updates, investigations reveal that it does not inherently guarantee worst-case differential privacy without additional noise, as the aggregated model can still leak information through mutual information bounds [218]. Theoretical proofs indicate that SA can only provide DP guarantees without added noise if the randomness within the aggregated update follows a Gaussian distribution with a non-singular covariance matrix, a condition rarely met in practice [79]. Consequently, hybrid frameworks combining cryptographic techniques with differential privacy, such as distributed matrix mechanisms, have been developed to achieve the utility of centralized DP with the privacy guarantees of distributed settings [90]. Moreover, the scope of theoretical analysis has expanded to include decentralized and multi-tier architectures, where privacy loss composition must account for social network effects and hierarchical trust models [110][255]. These developments illustrate a shift from static noise injection to dynamic, context-aware mechanisms that rigorously account for data heterogeneity, social externalities, and complex trust architectures.

Classification by Federated Learning Scenarios

The integration of Differential Privacy (DP) into Federated Learning (FL) necessitates a nuanced understanding of data partitioning strategies, as the privacy-utility trade-off varies significantly across Horizontal, Vertical, and Transfer learning scenarios. In Horizontal Federated Learning (HFL), where clients share the same feature space but possess different samples, the primary challenge lies in mitigating the impact of noise on model convergence while defending against gradient leakage attacks. Conventional approaches such as DP-FedAvg often suffer from performance degradation due to the inconsistency between local and global objectives, particularly under data heterogeneity [58]. To address this, recent advancements have moved beyond uniform noise injection. For instance, FedLAP-DP proposes synthesizing small sets of samples to approximate local loss landscapes, allowing for record-level DP with improved utility compared to standard gradient-sharing schemes [58]. Furthermore, the issue of data heterogeneity in HFL is tackled by algorithms like DP-SCAFFOLD, which incorporates control variates to correct user drift, demonstrating superior performance over DP-FedAvg when local updates are large or data distribution is highly non-IID [43].

While HFL is well-studied, the application of DP in Vertical Federated Learning (VFL), where clients share the same samples but hold different features, presents distinct challenges regarding intermediate result leakage. In VFL, privacy risks are not limited to model weights but extend to the intermediate representations exchanged during collaborative training. Hybrid frameworks such as HDP-VFL have been developed to secure these intermediate results using DP alongside cryptographic protocols like Homomorphic Encryption, ensuring joint differential privacy with negligible cost to training time or accuracy [257]. Similarly, secure and privacy-preserving VFL frameworks distribute the aggregator role across multiple servers to perform secure multiparty computation, applying DP to the final released model to protect both input and output privacy [283]. The unique topology of VFL requires specialized adaptations because the correlation between features held by different parties can lead to inference attacks that standard HFL defenses may not adequately address.

Transfer Federated Learning introduces another layer of complexity, particularly when leveraging knowledge from heterogeneous source datasets to enhance a target dataset under strict privacy constraints. Federated Transfer Learning with Differential Privacy formulates a privacy model that sits between local and central DP, addressing the fundamental costs associated with data heterogeneity and knowledge transfer [10]. This scenario is critical for domains where data distributions vary wildly across sites, requiring mechanisms that can transfer utility without compromising the privacy guarantees of the source domains.

Beyond the partitioning strategy, the granularity of privacy protection—whether at the record, user, or subject level—further differentiates DP-FL methodologies. Most early works focused on record-level DP, yet in cross-silo settings, a single user's data may span multiple silos, necessitating user-level DP guarantees. Uldp-FL addresses this by employing per-user weighted clipping to ensure user-level DP in general cross-silo settings, a significant departure from group-privacy approaches [181]. Similarly, Subject Granular Differential Privacy in Federated Learning explores subject-level privacy where an individual's information is distributed across multiple federation users, proposing algorithms like LocalGroupDP and HiGradAvgDP to enforce these guarantees locally [46]. The distinction is vital; as noted in critical analyses, applying record-level DP to client updates in scenarios where all records belong to a single client (e.g., a hospital) may fail to protect the client's overall privacy, rendering the guarantee insignificant in certain contexts [178].

The interplay between privacy mechanisms and system architecture also varies by scenario. In HFL, techniques like Secure Aggregation (SA) are often combined with DP to provide stronger end-to-end guarantees, as seen in frameworks like DDP-SA which integrate Local DP with additive secret sharing [4]. However, the efficacy of SA alone is limited; theoretical bounds indicate that privacy leakage through aggregated models reduces linearly with the number of users, suggesting that SA must be complemented by DP for robust protection [218]. Conversely, in scenarios involving heterogeneous privacy requirements among clients, frameworks like IntraShuffler introduce privacy-aware shuffling to disrupt gradient structures that could otherwise be exploited by an honest-but-curious server, a risk particularly pronounced in Heterogeneous DP settings [65].

Despite these advancements, significant challenges remain in balancing privacy, utility, and communication efficiency across all scenarios. Adaptive mechanisms are increasingly necessary to manage these trade-offs dynamically. For

example, Adap DP-FL utilizes adaptive gradient clipping and decreasing noise schedules to account for gradient heterogeneity and convergence states, outperforming static noise injection methods [84]. Additionally, the practicality of DP in FL is often questioned due to the accumulation of privacy loss over multiple communication rounds; theoretical work suggests that tuning the number of local iterations can mitigate divergence caused by DP noise, although some mechanisms like Laplace-based DP may still struggle to converge without careful parameterization [215]. Ultimately, the classification of DP-FL methods by scenario reveals that while horizontal settings benefit from variance reduction and adaptive noise, vertical and transfer settings require hybrid cryptographic-DP solutions and specialized privacy formulations to handle feature splitting and domain shifts effectively.

Differentiation of DP Models: Central, Local, and Shuffle

The theoretical landscape of privacy-preserving Federated Learning (FL) is fundamentally defined by the trust assumptions placed upon the aggregation server and, more critically, the specific definition of neighboring datasets used to calculate privacy guarantees. Characterizing FL solely by the presence or absence of a trusted server is insufficient; rather, the distinction lies in the interplay between Central Differential Privacy (CDP), Local Differential Privacy (LDP), and the emerging Shuffle model, each offering distinct trade-offs between utility, communication efficiency, and robustness against adversarial servers [12]. In the Central DP model, a trusted curator aggregates raw or un-perturbed updates before applying noise, providing strong utility but requiring absolute trust in the server [1]. Conversely, Local DP mandates that clients perturb their data or gradients prior to transmission, eliminating the need for a trusted server but often resulting in significant utility degradation due to the accumulation of noise across high-dimensional parameter spaces [25]. The Shuffle model has emerged as a critical intermediate framework that leverages an anonymous shuffler to decouple data from user identities, thereby achieving privacy amplification that bridges the gap between the high accuracy of CDP and the strong trust assumptions of LDP [30].

The definition of neighboring datasets varies significantly across these models, directly influencing the magnitude of required noise and the resulting privacy-utility trade-off. In CDP, neighboring datasets typically differ by a single record within the global aggregate, allowing the noise scale to depend on the sensitivity of the aggregate function rather than individual contributions [190]. In contrast, LDP defines neighbors at the user level, where the presence or absence of an entire user's dataset must be indistinguishable, necessitating noise levels that scale with the dimensionality of the model updates [297]. This dimension dependency poses a severe challenge in deep learning contexts, where high-dimensional gradients require excessive noise to satisfy LDP, often rendering the model useless without specialized mitigation strategies [158]. To mitigate this, recent works propose dimension selection techniques, such as privately selecting top-k gradient dimensions, to reduce the effective sensitivity and improve utility without compromising the LDP guarantee [297]. Furthermore, the shuffle model redefines the privacy landscape by introducing a trusted shuffler (or a secure multi-party computation equivalent) that permutes messages before they reach the analyzer [210]. This process amplifies privacy because the analyzer only sees a shuffled batch of messages, making it difficult to link specific updates to specific users, thus allowing for less noise to be added locally while maintaining the same global privacy guarantee [30].

Privacy amplification via shuffling represents a pivotal theoretical advancement, enabling protocols to achieve central-model accuracy without a fully trusted server. The amplification effect relies on the randomness introduced by the shuffling process, which effectively hides individual contributions within a crowd [261]. Protocols like SS-Topk leverage this by combining gradient sparsification with shuffling to handle scenarios where the model size exceeds the user population, demonstrating accuracy improvements of over 60% compared to standard LDP-based FL [30]. Similarly, the Camel framework utilizes secret-shared shuffling to ensure malicious security and integrity checks, deriving tighter bounds on privacy loss through Rényi Differential Privacy (RDP) analysis [123]. However, the applicability of shuffling is not universal; it faces compatibility issues in settings with Heterogeneous Differential Privacy (HDP), where clients select individual privacy budgets [65]. In such cases, standard shuffling conflicts with ϵ -aware aggregation strategies required for convergence, necessitating novel middleware defenses like IntraShuffler that perform parameter-level shuffling within privacy-compatible buckets to disrupt gradient structures while preserving aggregation weights [65].

Despite the theoretical elegance of the shuffle model, practical implementations must address the reality of untrusted or semi-honest servers and the potential for inference attacks. While Secure Aggregation (SecAgg) prevents the server from viewing individual updates, it does not inherently provide worst-case differential privacy guarantees without additional noise, as information can still leak through the aggregated sum over multiple rounds [218]. Moreover, recent analyses suggest that SecAgg alone is insufficient against sophisticated reconstruction attacks involving Sybil devices, reinforcing the necessity of combining cryptographic primitives with local noise injection [Reconstructing Individual Data Points in Federated Learning Hardened with Differential Privacy and Secure Aggregation]. The Inter-Silo Record-Level Differential Privacy (ISRL-DP) model further refines these distinctions for cross-silo settings, assuming trust within a silo (e.g., a hospital) but not in the server or other silos, effectively sitting between the extremes of CDP and LDP [190]. This nuanced trust model allows for optimal convergence rates comparable to CDP while adhering to

more realistic industrial trust assumptions [190]. Ultimately, the choice between CDP, LDP, and the shuffle model depends on the specific threat model, the dimensionality of the data, and the acceptable level of trust in the infrastructure, with the shuffle model increasingly viewed as the optimal balance for achieving high utility without relying on a fully trusted curator [1].

Privacy Accounting and Composition Mechanisms

The rigorous quantification of privacy loss over multiple training rounds constitutes a foundational challenge in differentially private federated learning, particularly when balancing strict privacy budgets against model utility. Traditional approaches often rely on standard composition theorems, which can result in overly conservative bounds that severely limit the number of feasible training iterations. To address this, advanced accounting frameworks such as Rényi Differential Privacy, $\mathcal{F}$ -Differential Privacy, and zero-concentrated DP have been developed to provide tighter bounds and more accurate tracking of cumulative privacy leakage. These frameworks are essential for enabling feasible training in decentralized settings where communication rounds are numerous and privacy constraints are stringent. [Federated $\mathcal{F}$ -Differential Privacy] [6] [20]

Standard composition mechanisms typically assume a uniform expenditure of the privacy budget across all learning rounds, an assumption that fails to account for the dynamic nature of model convergence. Research indicates that early training rounds often capture coarse-grained features that are robust to higher noise levels, whereas later rounds require higher signal-to-noise ratios to learn fine-grained details. Consequently, uniform budget allocation can lead to suboptimal utility. [174] proposes a time-adaptive framework that allows clients to expend their privacy budgets non-uniformly, saving budget in early rounds to spend more heavily in later stages where accuracy gains are most critical. This approach theoretically proves utility improvements, particularly for clients with stricter privacy constraints, by aligning noise injection with the learning trajectory. Similarly, [59] explores adaptive privacy budgets combined with adaptive clipping strategies, demonstrating that dynamically updating gradient sensitivity rather than using fixed values helps maintain model accuracy while preserving privacy throughout the training process. Further enhancing this adaptivity, [84] conducts adaptive gradient clipping for different clients and rounds while adding decreasing noises accordingly to address gradient heterogeneity and convergence. [35] also introduces a scoring function based on accuracy, loss, and training rounds to adjust the privacy budget dynamically.

The complexity of privacy accounting increases significantly in decentralized environments where the absence of a central trusted aggregator necessitates distributed noise injection and complex communication patterns. In such settings, accurately quantifying the privacy budget is challenging due to the interplay between local updates, decentralized communication topologies, and correlated noise. [6] addresses this by introducing accounting methods tailored to decentralized algorithms, specifically Pairwise Network $\mathcal{F}$ -DP and Secret-based $\mathcal{F}$ -Local DP. By leveraging $\mathcal{F}$ -DP theory and Markov chain concentration, this framework captures privacy amplification arising from sparse communication, local iterations, and correlated noise, yielding consistently tighter (ϵ, δ) bounds compared to RDP-based approaches. Furthermore, [Federated $\mathcal{F}$ -Differential Privacy] introduces a generic framework, PriFedSync, which operates on the record level and utilizes Gaussian differential privacy to provide robust guarantees against adversaries, empirically demonstrating superior trade-offs in computer vision tasks. [157] further improves upon prior analyses by reducing the Rényi DP parameter growth from quadratic to linear with respect to the number of training rounds, a significant theoretical advancement for long-running decentralized protocols.

While theoretical bounds provide worst-case guarantees, empirical measurements suggest that these bounds may be overly conservative in practice. [86] highlights that standard DP threat models can overestimate information leakage. By employing "canaries"-carefully crafted samples designed by a strong adversary to evaluate empirical privacy, the authors demonstrate that the actual per-round epsilon in realistic FL scenarios can be 4 to 5 times lower than the theoretical bound. This discrepancy suggests that advanced accounting mechanisms could potentially allow for more aggressive utility optimization if calibrated against empirical risk rather than solely relying on worst-case theoretical composition. [229] also introduces empirical privacy tests to interpret LDP levels more intuitively by instantiating various adversaries, confirming that worst-case attacks are often unrealistic in FL. Conversely, [178] warns against the misuse of relaxed DP definitions, such as (ϵ, δ) -DP with large δ values, arguing that such relaxations can undermine privacy guarantees in continuous data collection settings like FL, where sequential composition causes the effective ϵ to grow rapidly with the number of epochs.

To mitigate the accumulation of privacy loss without sacrificing utility, recent works have explored mechanisms that leverage inherent randomness or structural properties of the learning process. [79] investigates conditions under which Secure Aggregation can provide DP guarantees without explicit noise addition, proving that while theoretically possible under specific Gaussian randomness conditions, these are unlikely to hold in practice, thus necessitating additional

noise. However, the inherent randomness can still be leveraged to reduce the magnitude of added noise. In a similar vein, [238] proposes LPPA, a method that injects noise differences between sent and received gradients, utilizing noise flow conservation to ensure global elimination of noise impact, thereby achieving lossless model accuracy while enhancing privacy capacity beyond standard DP methods. [205] presents a novel approach where randomized quantization alone achieves Renyi DP guarantees, eliminating the need for explicit discrete noise and offering improved privacy-accuracy trade-offs through a two-level randomization process. These diverse strategies collectively illustrate that moving beyond standard composition theorems to adaptive, heterogeneous, and empirically validated accounting frameworks is essential for the practical deployment of privacy-preserving federated learning. [196] provides a theoretical proof for the optimal number of local steps and communication rounds that maximize convergence within a given privacy budget, revealing a direct correlation between these variables and the DP budget. [215] derives convergence rates for DP-based FedAvg, showing that tuning the number of local iterations can prohibit divergence caused by Laplace mechanisms and ensure convergence for Gaussian mechanisms.

Formal Security Definitions and Attacker Advantages

Formalizing security in dynamic distributed systems requires a rigorous transition from heuristic privacy measures to mathematically provable guarantees that explicitly quantify the attacker's advantage. A foundational element of this formalization is establishing the fundamental costs associated with these guarantees, particularly regarding communication overhead, which defines the feasible region for secure protocols. Research establishes an information-theoretic model demonstrating that secure aggregation imposes a fundamental communication cost of $\Omega(n \log d)$ bits per user, a figure significantly higher than non-secure transmission but lower than naive secure schemes [160]. This work highlights that achieving optimal accuracy-privacy trade-offs requires careful construction of linear schemes based on noisy sketches, challenging the assumption that security can be added without substantial overhead. Further characterizing this cost, other studies show that sparse random projections can significantly reduce the communication burden required to achieve central differential privacy levels while maintaining optimal accuracy, effectively specifying the fundamental price of using secure aggregation in differentially private federated learning [252].

The definition of attacker advantage is further complicated by specific threat models inherent to Federated Learning (FL), such as Data Leakage from Gradients (DLG) and model inversion attacks, which require formal notions of differential privacy to be directly related to the adversary's success probability. To address the lack of rigorous evaluation foundations, recent work extends formal notions of differential privacy to relate directly to attacker advantage within machine-supported verification frameworks like the Isabelle Insider and Infrastructure framework [14]. This formalization is critical for dynamic systems like satellite swarms, where the verification of security properties must be automated to ensure robustness against evolving attack vectors. Despite these formal efforts, a significant divergence exists between theoretical worst-case bounds and empirical reality, prompting the development of methods to measure actual per-round privacy loss. A novel methodology introduces the use of "canaries"-carefully crafted samples designed by a strong adversary-to evaluate empirical privacy in realistic scenarios, demonstrating that the empirical per-round epsilon is often four to five times lower than the theoretical bound [86]. Complementing this, other studies propose empirical privacy tests by instantiating multiple adversaries to measure the lower bounds of Local Differential Privacy, finding that the worst-case attacks assumed in theoretical bounds are often unrealistic in practice [229].

However, this empirical optimism is countered by demonstrations that even hardened protocols combining Distributed Differential Privacy and Secure Aggregation are vulnerable to reconstruction attacks via Sybil devices, exposing a power imbalance where the server orchestrates the protocol to expose individual user data [Reconstructing Individual Data Points in Federated Learning Hardened with Differential Privacy and Secure Aggregation]. The interplay between cryptographic primitives and privacy definitions further refines the understanding of attacker advantage, particularly regarding the sufficiency of Secure Aggregation alone. Investigations into conditions under which Secure Aggregation might provide differential privacy guarantees without added noise conclude that while theoretically possible under specific Gaussian randomness conditions, these are unlikely to hold in practice, thus necessitating explicit noise addition to bound the attacker's advantage [79]. Furthermore, while Secure Aggregation prevents the server from accessing individual updates, it does not inherently guarantee that information about individual datasets cannot leak through the aggregated model, with theoretical bounds indicating that privacy leakage reduces linearly with the number of users yet still leaving a residual advantage for sophisticated adversaries [218].

The limitations of differential privacy are also critically examined in contexts where extending it beyond interactive queries to record-level data release can lead to meaningless privacy guarantees due to sequential composition and large epsilon values, thereby creating a false sense of security [178]. Furthermore, integrating robustness against Byzantine attacks with privacy preservation introduces complex trade-offs; while noise added for privacy can inadvertently enhance robustness, tight analysis exhibits a fundamental trilemma proving that any algorithm ensuring both robustness and differential privacy incurs a specific error cost, thereby defining a hard limit on the utility an adversary can disrupt while remaining undetected [254]. Emerging architectures like Trusted Execution Environments aim to close the gap between theory and practice by ensuring server-side confidentiality, yet warnings persist that side-channels can circumvent differential privacy guarantees, underscoring the necessity of holistic security definitions that account for all potential avenues of attacker advantage in dynamic distributed environments [203].

Algorithmic Innovations and Noise Injection Strategies

The integration of Differential Privacy (DP) into Federated Learning (FL) introduces a fundamental tension between rigorous privacy guarantees and model utility, primarily due to the degradation caused by noise injection and gradient clipping. Recent algorithmic innovations have moved beyond uniform noise application toward strategies that adapt to the optimization landscape, gradient geometry, and training dynamics. A significant advancement involves modifying the optimization objective to enhance robustness against perturbations. Standard DP methods often result in sharp loss landscapes that are highly sensitive to weight perturbations, leading to severe performance degradation. To counter this, researchers have proposed integrating Sharpness Aware Minimization (SAM) into the FL process, which encourages the discovery of flat loss landscapes and reduces the norm of local updates, thereby diminishing the relative impact of DP noise [217]. This approach can be further refined by combining flatness-aware optimization with sparsification techniques to reduce the magnitude of random noise required for a given privacy level [217]. Additionally, immersion and invariance-based coding treats optimization algorithms as dynamical systems, embedding model parameters into a higher-dimensional system to offer desired DP levels while maintaining standard convergence rates [33].

Complementing optimization modifications, adaptive noise injection mechanisms have emerged as a critical strategy for balancing privacy and performance throughout the training lifecycle. Static noise levels are often suboptimal, as they may hinder early convergence or provide excessive privacy in later stages. To address this, time-varying noise amplitudes based on geometric series have been introduced, with analytical derivations showing their dependence on global aggregations to prevent premature convergence [293]. Furthermore, adaptive schemes that adjust noise based on gradient heterogeneity and convergence progress have demonstrated superior performance over uniform methods. Specifically, adaptive gradient clipping for different clients and rounds, combined with decreasing noise schedules, effectively mitigates the negative impact of heterogeneity [84]. Other approaches utilize scoring functions incorporating accuracy, loss, and cosine similarity to dynamically adjust privacy budgets, significantly reducing the total budget consumption without compromising accuracy [35]. In personalized FL settings, these adaptive concepts extend to analyzing trade-offs between privacy, convergence, and fairness, deriving optimal aggregation numbers to maximize performance under privacy constraints [93]. Priority-based approaches also inject noise based on feature importance, adding more perturbation to less critical parameters to preserve utility [227].

Structural modifications to the gradient updates themselves offer another potent method for noise mitigation by reducing the sensitivity of updates. Sparsification and dimension selection techniques allow for less noise injection for the same privacy guarantee. Federated learning schemes that sparsify local models before perturbation have proven effective in reducing necessary noise intensity while maintaining client-level DP [281]. Taking this further, frameworks that privately select top-k dimensions based on gradient contribution effectively relieve the dimension dependency problem where noise scales with model size [297]. Additionally, second-order methods combined with sparsification have been leveraged to achieve lower iteration complexity and amplified privacy guarantees [24]. Layer-wise adaptive noise injection mechanisms also optimize privacy protection by leveraging the insight that neural network layers contribute unevenly to model utility, dynamically injecting noise based on privacy sensitivity [22]. Quantization serves a dual purpose; stochastic quantization with mixed truncated geometric distributions can introduce necessary randomness for DP without additive noise, while discrete Gaussian mechanisms unify privacy and communication efficiency [125; 18].

The handling of gradient heterogeneity in large-scale models, particularly in domains like speech recognition, requires specialized clipping and normalization strategies. Research establishes that per-layer clipping combined with layer-wise gradient normalization is essential for mitigating clipping bias in deep transformer models, enabling viable DP-FL for automatic speech recognition with minimal word error rate degradation [234]. This contrasts with traditional uniform clipping, which often fails in deep architectures. Furthermore, algorithms augmented with control variates to correct user drift in heterogeneous data settings show superior performance over standard baselines when local updates are frequent or data heterogeneity is high [43].

Theoretical analyses regarding the number of iterations and communication rounds provide foundational guidelines for practical deployment. Formal derivations of convergence rates reveal that while Laplace-based DP-FedAvg may diverge, setting an optimal number of local iterations can prohibit divergence, whereas Gaussian mechanisms can converge to a constant error [215]. Investigations into the optimal balance between server queries and client replies

further demonstrate that properly setting these numbers can significantly improve final model accuracy [156]. In wireless environments, the physical layer offers unique opportunities for noise management. Studies demonstrate that inherent channel noise can be harnessed for privacy, with adaptive power control allowing for a trade-off between signal-to-noise ratio and privacy levels, effectively achieving privacy "for free" under certain thresholds [130]. Similarly, transmit power control across clients can adjust received signal levels to control noise perturbation effectively [195]. Reconfigurable intelligent surfaces (RIS) also boost receive signal power to improve the privacy-accuracy trade-off in over-the-air systems [148].

Advanced mathematical frameworks also contribute to more efficient noise accounting and injection. Proposals to harmonize various randomization mechanisms using Re ϵ -DP reduce reliance on Gaussian noise and boost convergence [81]. In decentralized settings, ϵ -DP is utilized to capture privacy amplification from sparse communication and correlated noise, yielding tighter bounds than Re ϵ -DP approaches [6]. To address incompatibility between real-valued DP noise and secure multiparty computation, Skellam mixture mechanisms have been proposed to reduce noise levels while ensuring compatibility with integer-based protocols [116]. Finally, mechanisms that inject noise differences which globally cancel out theoretically ensure lossless model accuracy while providing privacy protection superior to standard DP methods [238].

Adaptive and Layer-wise Noise Mechanisms

The evolution of differential privacy in federated learning has transitioned from static, uniform noise injection toward sophisticated, adaptive mechanisms designed to optimize the privacy-utility trade-off. Traditional approaches often apply a fixed privacy budget and uniform noise distribution across all model parameters, which can severely degrade model utility, particularly in deep neural networks where layers contribute unevenly to feature extraction. To address these inefficiencies, recent research has introduced layer-wise adaptive mechanisms that quantify privacy leakage per layer and dynamically adjust noise scales based on training progress and data heterogeneity. A primary advancement is the recognition that neural network layers possess varying degrees of sensitivity and importance to model performance. [22] proposes a novel framework leveraging the insight that layers contribute unevenly to utility and that privacy leakage can be quantified via Kullback-Leibler divergence between local and global model distributions. By dynamically injecting noise only into selected layers based on their privacy sensitivity, [22] demonstrates a substantial reduction in noise injection while improving accuracy compared to state-of-the-art methods. This layer-specific approach contrasts with uniform perturbation strategies, offering more granular control over the privacy budget. Similarly, [268] investigates the impact of noise placement in split learning architectures, revealing that introducing noise in later local layers offers a more favorable balance between accuracy and privacy, suggesting that the structural position of noise injection is critical for mitigating model inversion attacks without sacrificing early feature extraction capabilities.

Beyond spatial adaptation across layers, temporal adaptation of the privacy budget throughout the training lifecycle has emerged as a crucial strategy. The assumption that clients should expend their privacy budgets uniformly over time is challenged by the observation that early training rounds focus on coarse-grained features learnable at lower signal-to-noise ratios, whereas later rounds refine fine-grained features requiring higher precision. [174] proposes a time-adaptive framework allowing clients to conserve privacy budgets in early rounds to expend more in later stages, theoretically proving utility improvements for clients with strict privacy constraints. This dynamic allocation is further refined in [202], which utilizes Gaussian process regression and Contextual Multi-Armed Bandits to strategically allocate budgets per round, predicting accuracy changes to maximize long-term performance. Complementing these approaches, [84] introduces decreasing noise schedules aligned with gradient convergence, arguing that as the model stabilizes, the magnitude of added noise should diminish to prevent convergence stagnation. Furthermore, [35] introduces a scoring function that adjusts the privacy budget based on multiple factors including accuracy, loss, training rounds, and dataset characteristics, ensuring that budget adjustments remain within reasonable boundaries to prevent non-convergence.

The heterogeneity of client data and resources further necessitates adaptive mechanisms that move beyond one-size-fits-all noise policies. [227] suggests prioritizing features based on their relative importance, injecting more noise into less important parameters to preserve the integrity of critical model weights. This concept is expanded in [120], which addresses scenarios where clients possess varying privacy requirements, proposing a privacy-aware selection strategy to minimize training error by distinguishing between high-quality updates and those introducing substantial noise due to strict privacy constraints. Explainability in noise allocation represents another frontier, aiming to clarify the relationship between dynamic noise adjustments and privacy requirements. [235] utilizes Shapley values to assess the contribution of private attributes, guiding noise injection based on quantifiable privacy contributions rather than heuristic factors. Similarly, [70] combines HyperDimensional computing with a transparent noise accountant to track cumulative perturbations, ensuring clients add only the minimal incremental noise required.

While many methods rely on adding artificial noise, some research explores leveraging inherent system properties or alternative mathematical structures to achieve privacy with minimal utility loss. [147] presents a framework that applies a Noise-Free Differential Privacy mechanism within a federated model distillation context, eliminating the need for random noise addition while maintaining privacy guarantees. In wireless environments, [151] and [130] demonstrate how inherent channel noise and adaptive power control can be harnessed to provide "free" privacy, reducing the need for explicit noise injection. Additionally, [238] proposes a method where noise differences are injected and subsequently canceled out globally, ensuring lossless aggregation while protecting individual gradients. Despite these advancements, contradictions remain regarding the feasibility of noise-free approaches versus the robustness of calibrated noise; for instance, [79] argues that while secure aggregation can theoretically provide DP without added

noise under specific Gaussian conditions, such conditions are rarely met in practice, necessitating additional noise. Consequently, the field is moving toward hybrid models that combine layer-wise sensitivity, temporal budgeting, and explainable prioritization to navigate the complex landscape of privacy-preserving machine learning.

Optimization of Clipping and Sensitivity Control

The optimization of gradient clipping and sensitivity control serves as the foundational mechanism for bounding the influence of individual data points prior to noise injection in differentially private federated learning. Traditional approaches relying on fixed clipping norms often fail to accommodate the dynamic nature of model training, introducing significant bias or failing to adapt to gradient heterogeneity across clients. Recent literature has shifted decisively toward adaptive clipping strategies that dynamically adjust sensitivity bounds based on the convergence state and the statistical properties of local updates. For instance, adaptive clipping mechanisms within Gaussian mechanisms ensure that gradients are updated dynamically rather than relying on a static sensitivity threshold, demonstrating that such adaptivity, when paired with an adaptive privacy budget, can maintain model accuracy while preserving privacy in non-IID settings where gradient magnitudes vary significantly [59; 84]. This consensus is further reinforced by multi-objective optimization frameworks which posit that excessively high clipping values compromise privacy, while overly restrictive norms degrade performance; consequently, these frameworks dynamically adjust the clipping norm by integrating privacy and utility considerations into a unified optimization objective, theoretically proving convergence properties that outperform fixed-clipping baselines [88].

The challenge of selecting an optimal clipping norm becomes particularly acute in large-scale models, such as deep transformers used in automatic speech recognition, where standard global clipping techniques often fail due to severe layer-wise gradient heterogeneity. In these complex architectures, revisiting per-layer clipping and combining it with layer-wise adaptive gradient normalization has proven essential for mitigating clipping bias, allowing for strong privacy guarantees with minimal degradation in word error rates [234]. This granular approach highlights that sensitivity control must be context-aware, moving beyond simple global norms to strategies that respect the architectural depth and varying sensitivity of different model components. Furthermore, the integration of control variates offers a sophisticated complementary approach to clipping for correcting client drift, a phenomenon exacerbated in non-IID environments. Algorithms like DP-SCAFFOLD augment standard federated averaging with differential privacy by estimating and combining the global model update direction with local control variates; this effectively corrects user drift without degrading privacy levels, as the control variates depend only on stochastic gradients, thereby surpassing standard DP-FedAvg particularly when the number of local updates is large or data heterogeneity is high [43]. Building on this, adaptive extrapolation methods have been proposed to select global step sizes based on local update diversity, accelerating convergence without additional hyperparameters and addressing the slow convergence often observed in DP-FL due to client drift [109].

Beyond simple norm adjustment and drift correction, fairness-aware clipping strategies have emerged to address the disparate impact of privacy mechanisms on different client groups. These strategies employ clipping techniques specifically designed for biased model updates, utilizing Gaussian mechanisms to ensure that the global model remains fair despite the noise injection required for privacy, highlighting the tension where privacy requirements can inadvertently amplify biases present in local datasets if clipping is not managed carefully [87]. This trade-off is further analyzed in personalized settings, where frameworks optimize the number of global aggregations to balance convergence, privacy, and performance distribution fairness, revealing that joint optimization is feasible and necessary for equitable outcomes [93]. Alternative methods to traditional clipping involve sparsification and sketching to reduce sensitivity inherently, thereby lowering the magnitude of required noise. Techniques leveraging Sharpness Aware Minimization generate flat local models with smaller update norms, while local update sparsification further reduces sensitivity and improves robustness to DP noise [217]. Similarly, sparsified model perturbation schemes where local models are sparsified before Gaussian noise injection provide tight end-to-end privacy analysis confirming improved accuracy under client-level DP [281].

Contradictions and nuances exist regarding the efficacy of noise cancellation versus explicit clipping, suggesting that in decentralized settings, noise flow conservation might offer a viable alternative to strict sensitivity bounding. Lossless aggregation rules that inject noise differences to protect gradients while ensuring global noise elimination theoretically prove a privacy capacity $\sqrt{2}$ times greater than standard DP without accuracy loss, contrasting with methods that rely heavily on clipping which inherently introduce bias [238; 183]. However, practical constraints often necessitate careful clipping and noise calibration, as characterizations of the fundamental price of secure aggregation imply that while noise cancellation is theoretically appealing, achieving optimal accuracy under secure aggregation still requires

rigorous sensitivity management [252]. Additionally, quantization schemes shaped to follow Gaussian distributions can effectively achieve client-level DP through the quantization error itself, reducing the reliance on explicit clipping and separate noise addition [295]. These diverse approaches collectively underscore that optimizing clipping and sensitivity is not a one-size-fits-all solution but requires context-aware strategies that consider data heterogeneity, model architecture, and fairness objectives to navigate the complex privacy-utility trade-off.

Advanced Accounting Frameworks for Decentralized Settings

The transition from centralized to decentralized learning topologies necessitates a fundamental reimagining of privacy accounting mechanisms, as the absence of a trusted curator invalidates the assumptions underpinning traditional composition theorems. In centralized Federated Learning, privacy guarantees are often derived assuming a secure aggregation server that masks individual contributions before global summation. However, in fully decentralized settings where nodes communicate directly via gossip protocols or random walks, the lack of a central aggregator complicates the tracking of cumulative privacy loss across the network. Traditional accounting methods, such as those based on Rényi Differential Privacy, frequently yield overly conservative bounds in these environments because they fail to capture the specific structural advantages of decentralized communication, such as sparse connectivity and the potential for noise cancellation among neighbors. To address these limitations, recent research has introduced novel accounting frameworks tailored specifically for decentralized graphs, most notably Pairwise Network ϵ -DP and Secret-based ϵ -Local DP, which leverage the unique properties of network topology to achieve significantly tighter privacy bounds than their centralized counterparts [6].

The Pairwise Network ϵ -DP framework represents a significant advancement in quantifying privacy leakage between specific user pairs within a network, moving away from global worst-case assumptions. Unlike standard accounting methods that assume information propagates uniformly across the entire system, this framework utilizes tools from Markov chain concentration to model the random-walk nature of decentralized communication. This approach allows for the precise capture of privacy amplification arising from the sparsity of the communication graph and the number of local iterations performed between communication rounds. By focusing on the trajectory of information flow rather than static global sums, Pairwise Network ϵ -DP demonstrates that privacy leakage grows more slowly in sparse networks than previously estimated, effectively turning the communication bottleneck into a privacy feature. Empirical evaluations confirm that this method yields consistently tighter bounds compared to Rényi DP-based approaches, thereby enabling higher model utility for the same privacy budget [6]. This aligns with broader findings that full decentralization can inherently provide formal privacy gains, often matching the utility of the trusted curator model under specific network topologies like rings or complete graphs, a phenomenon formalized under the concept of Network DP [189].

Complementing the network-centric view, the Secret-based ϵ -Local DP framework introduces a mechanism for structured noise injection via shared secrets between connected users, directly addressing the threat of curious neighbors in a decentralized graph. In this paradigm, users exchange randomness seeds to generate pairwise-canceling correlated Gaussian noises, which are injected to protect local models at every communication round. Theoretically, this correlated noise cancels out during the aggregation phase, protecting the global model's accuracy while obscuring individual contributions from eavesdroppers and honest-but-curious peers. The Secret-based ϵ -Local DP accountant is specifically designed to handle the accumulation of non-canceling noise resulting from network sparsity, a critical factor often overlooked in simpler Local DP formulations. Research indicates that for arbitrary connected graphs, protocols utilizing such correlated noise can match the optimal privacy-utility trade-off of central DP, provided that every pair of connected users shares a secret hidden from others [292]. This concept of noise cancellation is further supported by protocols like Gopa, which demonstrate that exchanging pairwise-correlated noise terms along graph edges can nearly match the utility of a trusted curator, even when users communicate with only a logarithmic number of neighbors [213].

A critical distinction in these advanced frameworks is the rigorous treatment of noise correlation, contrasting with traditional Differential Privacy assumptions of independent noise addition. While some studies have suggested that inherent randomness in aggregated updates might suffice for privacy without added noise, rigorous analysis shows that such conditions are rarely met in practice, reinforcing the necessity of the structured, correlated noise mechanisms proposed in Secret-based ϵ -Local DP and Pairwise Network ϵ -DP [79]. The DECOR variant of decentralized SGD, for instance, theoretically proves that correlated noise injection allows the system to maintain the optimal privacy-utility frontier despite the lack of a central server, but this relies heavily on the assumption that secrets are securely shared and that the network remains sufficiently connected. If the graph is too sparse or secrets are compromised, the non-canceling noise components can accumulate, degrading utility. Consequently, the accounting frameworks must dynamically adjust to the graph's spectral properties and the density of secure links. Recent work on low-cost

decentralized learning, such as Zip-DL, corroborates these findings by showing that progressively neutralizing correlated noise can reduce membership inference attack success rates significantly while maintaining high accuracy, validating the practical viability of these sophisticated accounting methods [95].

The integration of these accounting methods into broader optimization strategies reveals important trade-offs between communication efficiency, convergence speed, and privacy guarantees. For example, while sparse communication enhances privacy amplification in Pairwise Network ϵ -DP, it may slow down convergence, requiring careful tuning of local iteration counts to balance the privacy-utility curve. Similarly, the overhead of establishing shared secrets for Secret-based ϵ -Local DP must be balanced against the privacy gains, particularly in dynamic networks where topology changes frequently. Moreover, the application of these frameworks extends beyond simple averaging to complex tasks like stochastic convex optimization and bandit problems, where the interplay between partial participation and correlated noise requires robust accounting to prevent privacy budget exhaustion [179]. Ultimately, the shift toward ϵ -DP-based accounting in decentralized settings marks a pivotal move from conservative, worst-case bounds to nuanced, topology-aware privacy guarantees that unlock the full potential of distributed learning architectures, allowing for algorithmic innovations that were previously deemed too privacy-costly under traditional accounting regimes.

Conclusion and Synthesis of Findings

The synthesis of current literature affirms that while Federated Learning establishes a foundational architecture for decentralized data processing, it is insufficient on its own to provide rigorous privacy guarantees against sophisticated inference and reconstruction attacks. Consequently, the integration of Differential Privacy has emerged as a critical necessity, transforming Federated Learning from a privacy-enhancing technique into a mathematically provable privacy-preserving framework [12] [1]. However, this integration introduces a complex triad of tensions between privacy, utility, and efficiency that defines the current state of the field. The consensus across multiple studies indicates that static noise addition, while theoretically sound, often results in unacceptable degradation of model accuracy, particularly in scenarios involving non-IID data distributions or small client datasets [28] [221].

To address the utility-privacy trade-off, the literature demonstrates a clear progression from uniform noise injection toward adaptive, hierarchical, and hybrid architectures. A significant body of work focuses on adaptive mechanisms that dynamically adjust privacy budgets based on training dynamics, such as loss convergence, gradient similarity, or specific training rounds. For instance, adaptive methods that modulate noise levels according to the cosine similarity between local and global models or the entropy of local datasets have shown promise in reducing the total privacy budget consumption while maintaining competitive accuracy [35] [172]. Furthermore, time-adaptive strategies that allocate larger privacy budgets to later training rounds, where fine-grained features are learned, have been proven to outperform uniform spending strategies [174] [53]. Layer-wise approaches further refine this by injecting noise only into sensitive layers, as demonstrated by methods that significantly reduce noise volume while improving accuracy [22].

Beyond temporal adaptation, structural innovations in how and where noise is applied are reshaping the landscape. Hierarchical Federated Learning introduces intermediate aggregation layers where calibrated noise can be added, effectively balancing the disparate utility-privacy trade-offs of central and local models [39] [255]. Similarly, the exploration of the shuffle model offers a middle ground, leveraging privacy amplification to achieve curator-model accuracy without requiring a trusted central server, although challenges remain in scaling these benefits to high-dimensional models [30] [65]. Additionally, novel approaches such as noise-free differential privacy via model distillation and lossless aggregation rules that utilize noise cancellation suggest that the paradigm of explicit noise addition may be evolving toward more subtle perturbation techniques [147] [238].

A critical and increasingly prominent finding is the intricate interplay between privacy mechanisms and algorithmic fairness. There is growing evidence that standard implementations can disproportionately degrade performance for underrepresented groups, thereby exacerbating existing biases within federated systems [49] [240]. This tension necessitates a shift from treating privacy and fairness as isolated constraints to viewing them as coupled objectives. Recent frameworks attempt to resolve this by incorporating fairness-aware gradient clipping or regularization terms directly into the optimization process, aiming to achieve a Pareto optimal balance among privacy, fairness, and utility [11] [256].

Despite these advancements, significant open challenges persist, particularly regarding communication efficiency and the heterogeneity of client requirements. The bandwidth constraints of mobile and edge devices demand solutions that simultaneously compress model updates and preserve privacy, leading to research into sparsification-amplified privacy and quantization techniques that integrate guarantees [233] [64] [242]. Moreover, the assumption of uniform privacy preferences across clients is increasingly recognized as unrealistic. Future advancements depend heavily on developing robust mechanisms for heterogeneous settings, where clients can select individual privacy budgets without compromising the global model's convergence or exposing structural patterns to inference attacks [122] [120]. Ultimately, the path forward requires resolving the tensions between these competing objectives through holistic system designs that are adaptive to data heterogeneity, communication constraints, and diverse ethical requirements [119] [2] [45].

Future Directions, Open Challenges, and Conclusion

The evolution of privacy-preserving machine learning has decisively shifted from the naive assumption that data localization ensures security to the recognition that rigorous mathematical guarantees are essential. While Federated Learning effectively mitigates the risks associated with central data repositories, it remains vulnerable to sophisticated inference and reconstruction attacks that exploit shared model updates. Consequently, the integration of Differential Privacy has become indispensable, transforming collaborative learning into a framework capable of withstanding adversaries with arbitrary background knowledge. However, this synthesis reveals that the mere combination of these paradigms is insufficient; the field is defined by a complex trilemma involving privacy guarantees, model utility, and communication efficiency. Static approaches to noise injection frequently result in unacceptable performance degradation, particularly in non-independent and identically distributed data settings, necessitating a move toward more nuanced, adaptive, and hybrid architectures.

Current research demonstrates a clear trajectory away from uniform privacy budgets and static clipping norms toward dynamic, context-aware mechanisms. Adaptive strategies that modulate noise levels based on training convergence, layer-wise sensitivity, and gradient heterogeneity have emerged as critical solutions for preserving model accuracy while maintaining robust privacy bounds. Furthermore, structural innovations, such as the shuffle model and sparsification-amplified privacy, leverage communication protocols and model compression to amplify privacy guarantees without proportional increases in noise. The integration of cryptographic primitives like Secure Aggregation and Multi-Party Computation with Differential Privacy offers a promising hybrid path, aiming to achieve the utility of central models without the trust assumptions of a centralized curator. Yet, these advancements introduce new challenges regarding computational overhead, scalability, and the verification of security properties in decentralized environments.

A pivotal insight from this review is the intricate and often adversarial relationship between privacy preservation and algorithmic fairness. Evidence suggests that standard differential privacy mechanisms can disproportionately degrade performance for underrepresented groups, thereby exacerbating existing biases rather than mitigating them. This tension underscores the necessity of treating privacy, fairness, and utility as coupled objectives rather than independent constraints. Future frameworks must incorporate fairness-aware optimization techniques, such as adaptive clipping and personalized budget allocation, to ensure that the pursuit of privacy does not come at the cost of equity. Additionally, the heterogeneity of client requirements demands personalized privacy mechanisms that accommodate varying sensitivity levels without compromising the global model's integrity or exposing structural patterns to inference attacks.

Despite significant progress, substantial gaps remain in achieving a holistic solution for next-generation distributed intelligence. The practical deployment of these systems is hindered by the lack of standardized benchmarking protocols, which often leads to inconsistent evaluations of the privacy-utility trade-off. Moreover, emerging frontiers such as federated unlearning, explainability, and the integration of quantum computing present new dimensions of complexity that current methods are only beginning to address. The path forward requires a unified approach that synthesizes adaptive noise mechanisms, robust cryptographic protocols, and fairness-aware optimization into coherent frameworks. Only by resolving the tensions between these competing objectives through rigorous, system-level design can the field realize the full potential of trustworthy, collaborative artificial intelligence.